conviction_tilt_max

equity strategy | family: equity-max | status: research | derived from `conviction_tilt`

What this strategy does

conviction_tilt with bigger bets for more return: base unit 0.007, 6x leverage cap

Why it should work. Concentrate capital where the surprise is largest; neutralize size so it is not a hidden bet.

Design notes (from the strategy's source file).

Max-size variants ("<name>_max") of the hand-built equity strategies.

Same signal, trim, tilt and neutralization as each base strategy; only the two dials that set how big and how levered the book is change (base_unit_fraction, gross-leverage cap = 6x). The multiplier for each was picked by sweeping 1.5x..12x the original base unit and taking the highest return subject to Sharpe >= 0.6 and max drawdown >= -40% (a worse drawdown is a fund-ending event, not a risk preference). The untouched 5%-of-ADV liquidity cap remains the real capacity constraint, and bites roughly four times as often at this sizing.

These are generated from the base strategy classes, so a change to a base strategy's weights flows through automatically.

parameter	value
base_unit_fraction	0.007
max_gross_leverage	6.0
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	priority
exposure_cap	open_book
beta_overlay	0.0

Performance

metric	conviction tilt max	market (CRSP VW)
Annualized return	9.26%	9.94%
Annualized volatility	12.65%	19.35%
Sharpe (NAV-pct)	0.76	0.59
Sortino	1.09	
Max drawdown	-40.5%	-55.5%
Calmar	0.23	
Final NAV ($10M start)	$56,133,710	
Worst day	-7.25%	
% positive days	49.8%	
Sharpe 1996-2001	1.08	

Sharpe 2002-2007	1.45	
Sharpe 2008-2013	-0.05	

Correlation of daily returns with the market: **0.04**.

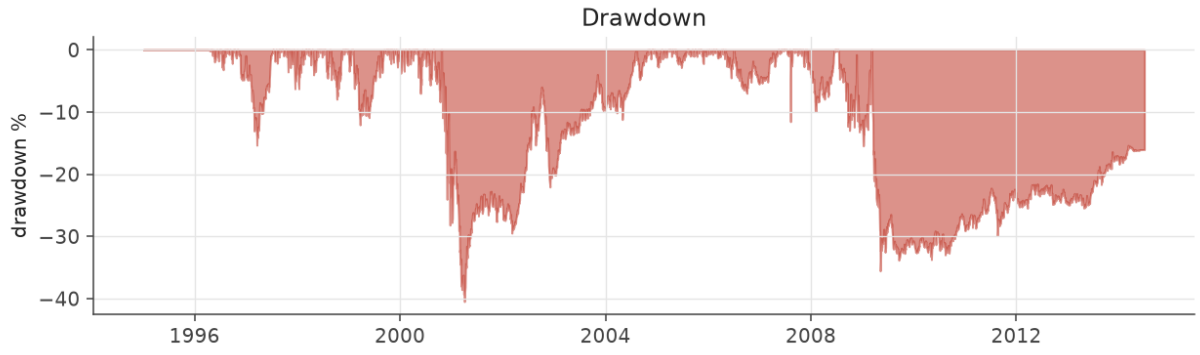
• best Sharpe among equity: `calm_rank_tilt` (1.47); this one: 0.764

• best return among equity: `market_boost_max` (0.116); this one: 0.0926

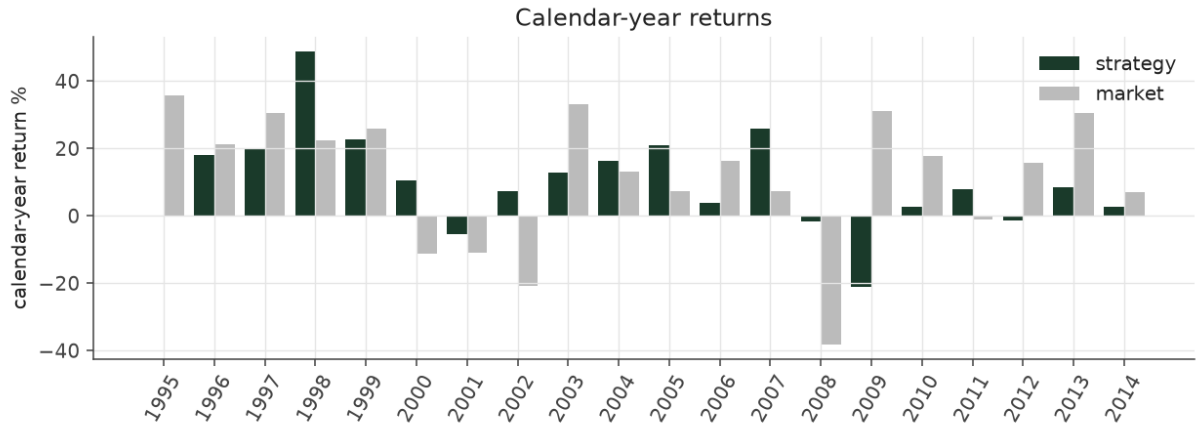
• best Calmar among equity: `calm_rank_tilt` (0.442); this one: 0.229



NAV (log scale)



Drawdown from running peak

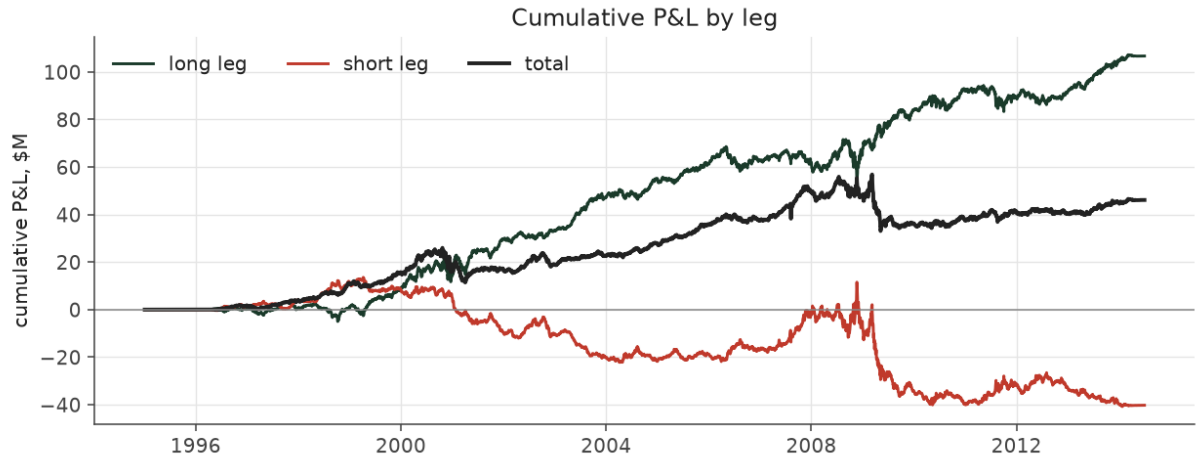


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	106.73	-40.24
P&L after costs ($M)	96.86	-50.72
of which common (average-event) component ($M)	87.76	-89.04

of which excess / PEAD component ($M)	18.97	48.80
excess minus costs ($M)	9.10	38.32
gross return per $ traded (bp)	191.03	-70.48
positions in the money (hit rate)	0.56	0.48
stand-alone Sharpe (leg only)	0.91	-0.23

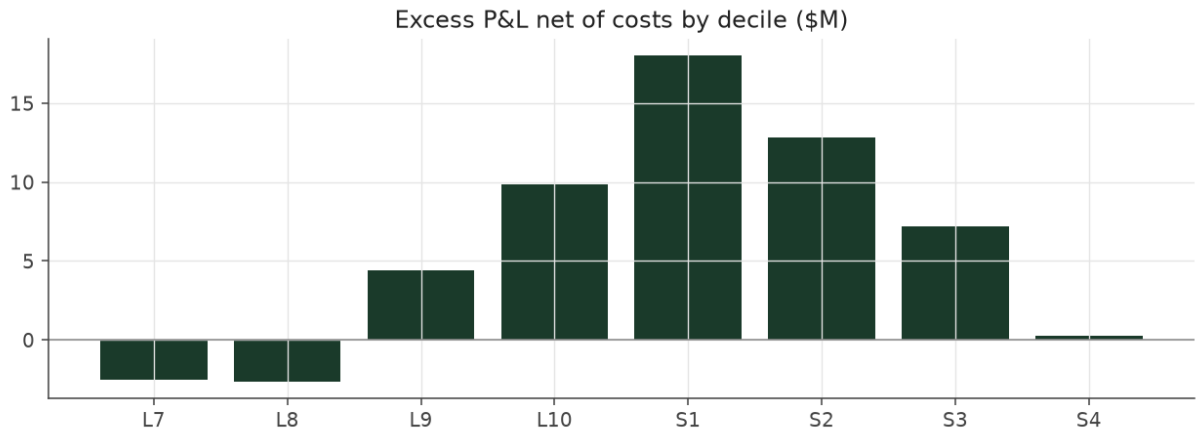
Daily P&L correlation between the two legs: **-0.73**.

• **The short leg loses money in absolute terms** (-40.2M before costs) while the long leg earns +106.7M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 87.8M / 89.0M. Stripping that common component, the surprise signal itself earned +19.0M on the long side and +48.8M on the short side.

• The two legs' daily P&L are -0.73 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 0.91 long / -0.23 short vs 0.76 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	7	6,507	350.56	3.01	-1.84	0.66	-2.50	-5.3%	-71.27
long	8	16,600	1,189.48	17.66	-0.49	2.17	-2.66	-5.6%	-22.39
long	9	18,545	1,711.44	29.11	7.42	2.99	4.43	9.3%	25.91
long	10	20,109	2,335.43	47.08	13.88	4.05	9.83	20.7%	42.08
short	1	18,252	2,270.25	-19.32	22.21	4.19	18.02	38.0%	79.39
short	2	18,869	1,989.82	-18.24	16.46	3.62	12.84	27.1%	64.53
short	3	15,351	1,305.57	-11.36	9.58	2.41	7.17	15.1%	54.91
short	4	2,369	144.14	-1.81	0.55	0.26	0.29	0.6%	20.03



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	13,888	581.47	-3.47	0.26	1.96	-1.70	-3.6%	-29.14
2	22,218	1,955.10	17.75	23.86	5.03	18.83	39.7%	96.32

3	25,033	2,760.28	17.52	21.96	5.50	16.46	34.7%	59.62
4	27,193	2,976.82	8.71	12.79	4.53	8.26	17.4%	27.75
5	28,270	3,023.01	5.62	8.88	3.32	5.57	11.7%	18.41

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	13,205	292.93	2.77	4.75	1.47	3.29	6.9%	112.14
2	22,552	1,802.69	8.17	16.26	5.41	10.85	22.9%	60.20
3	25,317	3,027.36	19.64	27.26	6.05	21.21	44.7%	70.05
4	27,157	3,113.17	7.07	10.93	4.36	6.58	13.9%	21.12
5	28,366	3,059.68	8.42	8.46	3.06	5.40	11.4%	17.66

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	41,721	3,891.28	-1.89	29.89	6.76	23.13	48.8%	59.44
41-60d	27,618	2,783.50	4.91	16.22	5.44	10.78	22.7%	38.72
61-90d	37,796	3,842.48	26.99	7.72	6.29	1.43	3.0%	3.72
>90d	9,467	779.43	16.12	13.94	1.85	12.08	25.5%	155.01

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	32,241	1,869.17	16.25	19.00	3.27	15.73	33.2%	84.16
2002-07	38,279	3,991.48	30.95	36.58	7.08	29.50	62.2%	73.89
2008-13	46,082	5,436.03	-1.06	12.20	10.00	2.19	4.6%	4.04

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	21,113	1,970.91	10.12	4.54	3.54	1.00	2.1%	5.09
Chems	2,548	253.96	0.79	1.68	0.44	1.24	2.6%	48.80
Durbl	2,821	260.62	0.26	1.56	0.48	1.07	2.3%	41.25
Enrgy	6,341	631.50	6.25	8.48	1.05	7.43	15.7%	117.72
Hlth	11,395	1,036.29	5.93	4.69	1.99	2.70	5.7%	26.04
Manuf	11,900	1,128.74	1.26	5.16	2.00	3.16	6.7%	28.00
Money	17,269	1,776.47	4.13	14.28	3.17	11.11	23.4%	62.53
NoDur	4,907	476.76	2.25	2.61	0.83	1.78	3.8%	37.32
Other	18,047	1,743.08	9.64	13.72	3.36	10.36	21.8%	59.44
Shops	11,570	1,114.70	6.81	8.08	2.01	6.06	12.8%	54.40

Telcm	3,954	391.27	-0.04	2.10	0.66	1.45	3.0%	36.92
Utils	4,737	512.38	-1.27	0.87	0.82	0.05	0.1%	1.02

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	39,014	3,976.12	4.41	12.98	6.56	6.42	13.5%	16.16
low vol	38,900	3,571.91	19.63	27.09	7.00	20.09	42.4%	56.26
mid vol	36,898	3,595.49	20.53	26.20	6.49	19.71	41.6%	54.82

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	9.26%	-0.00%
0.5%	5.6	8.67%	0.59%
1.0%	11.3	8.01%	1.25%
2.0%	22.6	6.42%	2.85%
5.0%	56.4	-100.00%	109.26%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.